PRD — Informal-Settlement Address & Emergency Dispatch Registry

Status: Draft v1, pending team/instructor review

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1. Problem

Informal settlements lack addresses. Existing efforts (Addressing the Unaddressed, Cape Town QR pilot) solve *initial* addressing only — one-time, static, built for postal/KYC. None close the loop when structures change, get demolished, or turn out unreachable. None route live emergency dispatch.

2. Goal

A living, offline-first registry that: - Assigns stable IDs to un-addressed structures - Builds a routable path-graph through informal footpaths - Stays current via feedback, not manual resurvey - Feeds a real dispatch backend - Keeps identity separate from location (consent-by-design)

3. Pilot Scope

Partial coverage. College campus, standing in for a settlement. Dispatch functions only inside mapped zone. No real-site access this term.

Important — open: which dispatch backend(s) this integrates with (municipal, NGO, both) is unresolved.

4. System Architecture

Component	Responsibility
Mapper app	Offline capture: pin, photo, breadcrumb, services
Structural registry	ID, location, footprint, status, services, last-verified. No PII
Consent vault	Resident identity. Separate store. Opt-in only
Path-graph service	Walkable network, built from merged breadcrumbs
Dedup/review queue	Human check on possible-duplicate structures at sync
Dispatch API	Code → location + route
Change-detection pipeline	Prioritizes re-survey zones from imagery
Grievance channel	In-app complaint, geo+photo tagged

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5. System Flow

1. **Bootstrap** — Candidate structure footprints pre-loaded from open building-footprint datasets, where available for the region.
2. **Map** — Mapper walks offline. Confirms/adjusts bootstrapped pins or drops new ones. Captures photo, material, service connections, GPS breadcrumb from road to door.
3. **ID assignment** — One stable code per structure, checked against the dedup queue before being finalized (see §6.1).
4. **Sync** — Opportunistic, whenever connectivity appears. Full offline capability otherwise.
5. **Path-graph build** — Breadcrumbs across mappers merged into one walkable graph (§6.2).
6. **Dispatch** — Dispatcher looks up code, gets location + route, sends responder.
7. **Feedback loop** — Failed dispatch or resident complaint both re-trigger verification. Same mechanism updates emergency and service data.
8. **Change detection** — Imagery diffing flags likely-changed zones for mapper revisit. Slow cadence, not real-time (§6.4).

6. Key Design Decisions

6.1 Structure identity & dedup

One code per physical structure. Code itself carries no location info — deterministic means *idempotent on revisit*, not geohash-derived. Idempotency is enforced, not assumed: on sync, any newly created structure within a tuned proximity radius of an already-synced one is flagged as a possible duplicate and routed to human review (paired photos, side by side) rather than auto-merged. No CRDT-style auto-merge for entity identity — CRDTs are used only for field-level edits to an *already-confirmed* structure (e.g. two mappers updating service data offline), not for deciding whether two pins are the same building.

Important — open: what counts as one “structure” for partitioned/shared dwellings is unresolved. This decision upstream-affects the dedup radius and the ID scheme.

6.2 Path-graph construction

Breadcrumbs merged via incremental clustering (k-means-style seed placement, weighted Delaunay triangulation for connectivity) rather than seed-and-spline map-matching, since the latter has no mechanism to suppress spurious paths from noisy traces — a known risk in dense, irregular alley layouts. Treated as its own scoped sub-milestone, validated on campus/synthetic traces before integration.

6.3 Consent

Identity opt-in, stored separately from structural data, by default. Dispatch does not carry occupant context unless resident opted in — accepted trade-off, since occupant context is normally only available after responders reach the scene anyway.

6.4 Imagery & re-verification cadence

Two-speed model, not one: - **Fast signal:** dispatch failure + resident complaint. Real-time. - **Slow signal:** imagery diffing. Weeks-to-months cadence, bounded by free-imagery refresh rates — not the few-day cadence commissioned capture would allow, which is out of scope (cost, drone logistics).

Free open building-footprint datasets (regional coverage varies) used as bootstrap layer and slow-cadence baseline, not as a live-change source.

6.5 Grievance channel

QR placard dropped as a required dependency; grievance reporting lives entirely in-app. Each report is geo- and photo-tagged and cross-checked against the structure's registered location. Rate-limited per device/account. Weighted by `verification_method` (dispatch-feedback-sourced updates outrank a single unverified resident report until corroborated).

6.6 Service-connection data

Stored as structured records per service (type, source, supply hours, quality, shared-with count, last-verified, verification method), not a binary flag. Same re-verification mechanism as structural status: full survey once, cheap "still true?" checks after.

Important — open: whether structure-level service-connection data (no identity attached) carries the same surveillance-trust risk as resident PII is unresolved.

6.7 AI scope

Used for: path-graph clustering, structure-footprint bootstrapping, change-detection prioritization, low-literacy capture assistance. Deliberately excluded from: ID assignment, sync conflict/identity resolution.

7. Open Questions

- Structure definition for partitioned/shared dwellings (§6.1)
- Dispatch backend(s) to integrate with (§3)
- Surveillance-trust risk of service-connection data absent identity (§6.6)

8. Known Risks

- Path-graph clustering accuracy under real GPS noise — unvalidated beyond synthetic/campus traces
- Open building-footprint model accuracy on informal structures specifically — historically the hardest case for these models
- Manual dedup review queue is a scaling bottleneck beyond pilot size
- Regional coverage gaps in free imagery/footprint datasets